



United International University

Department of Computer Science and Engineering

SOC 2101: Society, Environment and Engineering Ethics

Final Exam. : Summer 2024

Total Marks: 35 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the 5 questions. Numbers to the right of the questions denote their marks.

1. CleanEnv, an air purifier company, recently gained attention for its innovative air filter, which cleans air 45% faster than conventional HEPA filters and, unlike others on the market, is washable, extending its lifespan by 450%. However, John, an engineer at the company, discovered that the filter's air purification capacity decreased by 25% after each wash. When he raised the issue with management, they dismissed his concerns. Later, he learned that the company sends higher-quality, durable filters to social media influencers while selling cheaper, inferior versions to the public. After his promotion was withheld for undisclosed reasons, John believed it was due to his vocal criticism, prompting him to quit and expose the company's misconduct to the public. Which of the IEEE codes of ethics were violated and which were followed? Justify your answer. [7]
2. A healthcare company is developing a machine learning algorithm to assist in predicting the likelihood of patients developing certain chronic diseases based on their medical records, lifestyle data, and genetic information. The algorithm is expected to improve early diagnosis and help doctors provide timely interventions. However, concerns arise that the data used to train the model disproportionately represent patients from specific socioeconomic or racial groups, leading to biased predictions for other demographics.
 - i Among the following three biases -cognitive, statistical, and historical, which type of bias is more likely to occur in this scenario? [1]
 - ii What are the ethical implications of using biased data in training machine learning models for healthcare? Discuss the potential impact on patient care, trust in the healthcare system, and how algorithmic bias might disproportionately affect vulnerable populations. [4]
 - iii Propose strategies to mitigate bias and ensure that the model serves all patients fairly and accurately. [2]
3. At Photogram, one of the world's largest social media platforms, a controversial new project is about to launch. The company plans to analyze public posts to better understand user sentiment on sensitive topics, including mental health and political views. The leadership at Photogram believes this data could help governments and organizations address societal issues more effectively. However, concerns have emerged about the ethical implications of using personal data on sensitive topics without explicit consent from users, potentially breaching their trust and expectations of privacy.

The team, led by Emily, a senior researcher, is tasked with analyzing posts, comments, and hashtags from millions of public profiles. "These posts are public," Emily says confidently. "We're not invading anyone's privacy."

But as they prepare to roll out the data collection algorithms, James, a junior team member, raises a concern. "This is about mental health and personal beliefs. Just because people post publicly doesn't mean they expect us to analyze their deepest thoughts for corporate research. We need to be careful." Please point out the ethical concerns and what necessary steps should be taken? [7]
4. In a world on the brink of war, the small but powerful nation of Valenko faces a critical threat from surrounding hostile forces. With limited resources, Valenko's military leadership turns to a desperate measure: Project Ascend. The goal is to create an army of superhuman soldiers, capable of securing victory through the use of a revolutionary serum called V-23.

V-23 grants soldiers enhanced strength, speed, and reflexes for a limited period. However, the serum is not without its risks. Unbeknownst to the soldiers, V-23's side effects include potential organ failure, neurological damage, and even death. Desperate to gain an edge in the war, Valenko's leaders keep these dangers hidden from the participants, viewing them as necessary sacrifices for victory. Their plan is to gather enough data to refine the serum over time and, eventually, reduce the risks. Write down the data collection ethics with a consent letter. [7]
5. One day in Tony Stark's lab, Bruce Banner (The Hulk) and Tony Stark (Iron Man) are working on a groundbreaking project: finding a way to make Stark Industries' tech greener. With Stark's technological empire growing exponentially and generating huge amounts of electronic waste (e-waste), Banner warns that they are hurting the planet.

“I have been analyzing the data,” Banner says, pulling up a holographic display of Earth’s environment. “Your computers, devices, all that advanced tech—it’s leaving behind mountains of toxic waste and putting a strain on natural resources. It’s not just humans affected; the planet can’t take much more.”

Tony raises an eyebrow, slightly defensive. “You’re telling me that in trying to save the world, we’re slowly killing it? Come on, big guy. There has to be a way to balance innovation with sustainability.”

Bruce nods, eyes serious. “There is, but it is going to take some changes. We need to rethink how we are designing, manufacturing, and disposing of these devices.” Do you agree with Bruce’s statement regarding e-waste? Give support in favour of your answer. [7]